Univariate :

1) Central Dedency

[11]:

```
for colName in quan:
            descriptive[colName]["Mean"]= dataset[colName].mean()
            descriptive[colName]["Median"]=  dataset[colName].median()
            descriptive[colName]["Mode"]=  dataset[colName].mode()[0]
descriptive
```

[11]:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0

In this image, Means almost close for Sl_no, ssc_p, degress_p, etest_p, mba_p. but salary only having high level difference in mean and median. So, salary having outlier.

2) Percentile

```
for colName in quan:
            descriptive[colName]["Mean"]= dataset[colName].mean()
            descriptive[colName]["Median"]=  dataset[colName].median()
            descriptive[colName]["Mode"]=  dataset[colName].mode()[0]
            descriptive[colName]["Q1:25%"]=  dataset.describe()[colName]["25%"]
            descriptive[colName]["Q2:50%"]= dataset.describe()[colName]["50%"]
            descriptive[colName]["Q3:75%"]= dataset.describe()[colName]["75%"]
            descriptive[colName]["99%"]= np.percentile(dataset[colName],99)
            descriptive[colName]["Max"]=  dataset.describe()[colName]["max"]
descriptive
```

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
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Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Max	215.0	89.4	97.7	91.0	98.0	77.89	940000.0

```
import numpy as np
```

In this image, Q1, Q2, Q3, 99%, max are different type of location for splitting data percentage.

For Examples, Consider as ssc_p

Q1 = 25% , Q2= 50% = 60.6 -67.0 = overall 7% percentage difference from Q1 to Q2

Q2 = 50% , Q3= 75% = 67.0 -75.7 = overall 8% percentage difference from Q2 to Q3

Q3 = 50% , 99% = 75.7-87.0 = overall 12% percentage difference from Q3 to 99%

99% , Q4= 100% or MAX = 87.0 -89.4 = overall 2% percentage difference from 99% to 100%

Based on above data each percentage wise categorize the data percentage.

Find the IQR:

A) The InterQuartile range. Compare the interquartile ranges
B) Any outliers in either set.

The five number summary for the day and night class is

a. The interquartile range. Compare the two interquartile ranges.

b. Any outliers in either set.

The five number summary for the day and night classes is

	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

A) IQR = Q3-Q1

IQR for day = 82.5 -56 = 26.5
IQR for night = 89 – 78 =11

Difference between night and day IQR = 15.5

B) less Outlier = Q1-1.5 * IQR
greater Outlier = Q3+1.5 *IQR

Day outlier:
Less outlier = 56 -1.5*26.5 = 16.25
Greater Outlier = 89 +1.5* 26.5 = 128.75
Day – There is not less or greater outlier

Night outlier:
Less outlier = 78 -1.5 * 11 = 61.5
Greater Outlier = 89+1.5* 11 = 105.5
night – There is having less outlier because the minimum values is 25.5. But less outlier should be 61.50. it is 25.5 is less than 61.5. The greater outlier is not available in this night dataset.